Current makes a magnetic field

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Learning target: Connect current direction with the magnetic field around a wire.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 7, Day 9.*
6 A in each branch.

02. Worked example: explain the model
Counterclockwise tangent arrows; dot represents tip of an outward current arrow.

03. Read and interpret the model
Counterclockwise when viewed facing the page.

04. Apply the physics
It doubles.

05. Explain your reasoning
Field direction reverses; ideal magnitude stays the same.

06. Review: connect an earlier idea *Review: Unit 7, Day 1.*
Repulsion: each force points away from the other object.

07. Exit check
It halves.

08. STRETCH: test the model
Field circles reverse clockwise; circular geometry and equal-current field magnitude at equal distance stay the same.

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01. Warm-up: bring back an earlier idea *Review: Unit 7, Day 9.*
Parallel 4 Ω; total 10 Ω; supply current 2 A.

02. Worked example: explain the model
Counterclockwise tangent arrows; dot represents tip of an outward current arrow.

03. Read and interpret the model
Counterclockwise when viewed facing the page.

04. Apply the physics
5.0×10^{-5} T.

05. Explain your reasoning
Field direction reverses; ideal magnitude stays the same.

06. Review: connect an earlier idea *Review: Unit 7, Day 1.*
Polarization redistributes charge. Nearer opposite charge can attract more strongly than farther like charge repels; net charge may remain zero.

07. Exit check
3/2 or 1.5 times the original.

08. STRETCH: test the model
$B=10^{-5}$ T; at 0.20 m, 5×10^{-6} T, with direction set by current.